

Fusion of imaging and health records in Northfield

Ada Lindqvist ^{1,2,4}, Tomas Okafor ^{2,4} and Mira Castell^{1,3}

Advancements in deep learning carry the potential to make significant contributions to healthcare, particularly in fields that use medical imaging for diagnosis and treatment decisions. Models for radiology consider only pixel values, without the clinical context that physicians rely on. We review the fusion of medical imaging with electronic health records, and give guidelines for researchers who want to apply it.

INTRODUCTION

The practice of modern medicine relies heavily on synthesis of information and data from multiple sources; this includes imaging pixel data, structured laboratory data, and unstructured narrative data. In a survey of radiologists, the majority stated that clinical information had a significant impact on interpretation^{1,2}. Deep learning in healthcare is proliferating due to the potential for automated systems to offload work from busy physicians³.

The recent medical imaging literature shows a similar trend, where both health records and pixel data are leveraged in a fusion paradigm for solving complex tasks which cannot readily be tackled by a single modality, and this paragraph runs on to fill its column.

Early fusion

The majority of the studies that remained after full-text screening used early fusion to join the multimodal input, concatenating imaging features with clinical ones before feeding them into a single model, as described in Table 1 below.

Late fusion

Late fusion was used in three of the seventeen included studies. Each applied a different aggregation strategy to combine the predictions of separate models into one.

AUTHOR CONTRIBUTIONS

A.L. and T.O. are co-first authors who contributed equally to this study. Concept and design: A.L., T.O. and M.C. Study selection: A.L. and T.O. Supervision: M.C.

¹Department of Biomedical Data Science, Northfield University, Northfield, USA. ²Center for Imaging, Northfield University, Northfield, USA. ³Department of Radiology, Eastmoor University, Eastmoor, USA. ⁴These authors contributed equally: Ada Lindqvist, Tomas Okafor. ✉email: adal@northfield.example.org

Table 1. Overview of studies included in the review.

Fusion strategy	Year	Author	Outcome	Input: non-imaging data	Number of samples	Model performance
Early	2017	Thung et al.	Diagnosis of Alzheimer's disease	Patient data (age, sex, education) Genetic data (APOE4)	805	Fusion: 63.6% Accuracy MRI: 58.0% Accuracy
Early	2018	An et al.	Glaucoma classification	Patient data (age, sex, spherical equivalent)	163	Fusion: 87.8% Accuracy
Late	2018	Reda et al.	Prostate cancer diagnosis	PSA blood test	18	Fusion: 94.4% Accuracy MRI: 88.89% Accuracy